Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	LLM-Enabled Evidence-Grounded Decision-Support Platform for Rare Cancers			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (x**), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	정보통신		바이오/의료	
	Medium Category	Small Category	Medium Category	Small Category
	소프트웨어	SW솔루션(300802)	의료정보 및 시스템	기타 의료정보 및 시스템(500606)
Concept of Proposed Technology	An LLM-enabled, retrieval-augmented decision-support platform for rare cancers that are underrepresented in typical trials (e.g., salivary gland and sinonasal malignancies, as well as other rare histologies). The system continuously ingests and structures public literature, clinical-trial records, and drug/target knowledge into disease-focused RAG libraries and an evidence graph. Given a patient's diagnosis, prior treatments, and molecular profile, it generates a tumor board-style report with: (i) ranked investigational therapy and rational combination options with emphasis on novel/personalized strategies beyond routine standard targets, (ii) transparent citations and evidence grading, (iii) uncertainty handling including abstention when evidence is insufficient, and (iv) automated clinical-trial matching. A second output format produces cohort/target dossiers suitable for partnership discussions with pharma/biotech and for planning external pilots. **[System Architecture Overview]**			

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The platform is built as a **multi-layer Agentic RAG (Retrieval-Augmented Generation) system**, combining disease-specific retrieval, an evidence graph, constrained generation, and automated evaluation. Key architectural layers include:

1)Evidence Ingestion and Normalization Layer

- Continuous ingestion of public biomedical literature (PubMed/PMC), clinical trial registries (ClinicalTrials.gov), drug labels, and permitted curated oncology resources.
- Entity normalization across histology, molecular biomarkers, drugs, targets, and endpoints using controlled vocabularies and ontology mappings.
- Paragraph-level chunking with persistent provenance identifiers to enable traceability and downstream auditing.

2)Disease-Focused RAG Libraries and Evidence Graph.

- Construction of **disease-specific RAG libraries** for selected rare cancers, optimized for histology- and program-level retrieval rather than generic keyword search
- Standard actionable alterations (e.g., AR, HER2, and NTRK fusions) are incorporated as positive controls**,** but the system is explicitly designed for **post-standard personalization** when such targets are absent, exhausted, or not clinically applicable.
- Accordingly, retrieval and indexing are driven primarily by **program- and state-level molecular features**, including epigenetic dependency, DNA damage response/replication stress, immune phenotype, pathway-state activation, and resistance mechanisms.
- A complementary **evidence graph** encodes structured relationships among disease subtype, molecular programs, therapies, trials, and outcomes, and is used to support retrieval reranking, evidence consistency checks, and downstream prioritization logic.
- All retrieved evidence maintains paragraph-level provenance and evidence tagging, enabling transparent, auditable, and reproducible decision support in selected rare cancers.

3)Agentic Query Orchestration and Constrained Generation

- Patient- or cohort-level inputs (diagnosis, prior therapies, molecular profile, and clinical constraints) are expanded through hypothesis-driven query orchestration that explores relevant pathway states, resistance mechanisms, and trial eligibility variants.
- Generation is strictly constrained through fixed output schemas, mandatory citation requirements, and evidence-level checks. The system is explicitly designed to **abstain or defer to trial-first recommendations** when evidence thresholds are not met, rather than hallucinating low-confidence options.

4)Transparent Prioritization and Scoring Engine

- Candidate investigational therapies and rational combinations are ranked using a transparent, reproducible scoring framework that integrates:
    - mechanistic plausibility,
    - level and relevance of supporting evidence,
    - safety and pharmacologic considerations,
    - feasibility under patient-specific constraints, and
    - biomarker or assay suitability.
- All scores are traceable to underlying evidence nodes, enabling clinician review and regulatory inspection.

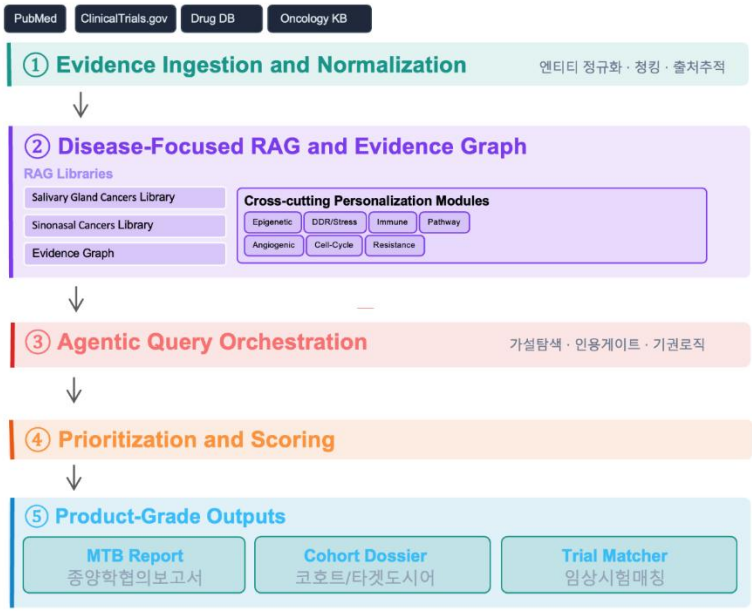
5)Product-Grade Outputs and Evaluation Harness

- The primary output is a **tumor board–style clinical decision-support report** presenting tiered investigational options, supporting evidence with citations, uncertainty flags, and matched clinical trials.
- A secondary output generates **cohort- and target-level dossiers** that summarize evidence landscapes, eligible patient populations, and trial concepts, directly supporting pharma/biotech collaboration and downstream business development.
- An integrated evaluation harness monitors grounding quality, abstention behavior, retrieval relevance, and clinical concordance in shadow-mode validation.

[Technical Differentiation]

The technology requires agentic orchestration, evidence governance, and the integration of real-world clinical data into product-grade LLM systems for biomedical decision support**.**

- **Agentic RAG design** that moves beyond single-pass retrieval to hypothesis-driven evidence exploration, while enabling alignment between public evidence and clinically curated real-world datasets.
- **Evidence governance by design**, including citation gates, evidence consistency checks, and explicit abstention logic suitable for high-stakes clinical contexts and clinical data–augmented decision making.
- **Evaluation-first development**, with built-in metrics for grounding, safety, and clinical concordance, supported by validation against institutional real-world cohorts rather than post-hoc assessment alone.
- **Product-ready architecture**, enabling versioning, monitoring, and deployment as reusable disease modules that can incorporate clinical feedback and real-world data updates, rather than one-off research prototypes.



Rationale for Joint R&D Project

Rare cancers are frequently under-represented or excluded from conventional trials, leaving clinicians with fragmented evidence (small cohorts, single-arm studies, and basket trials) and limited subtype-specific guidance—particularly when standard actionable alterations are absent, exhausted, or not applicable. A joint R&D project is needed because robust clinical translation requires both (1) an industry-grade AI platform (RAG infrastructure, evaluation harness, and productized report generation) and (2) clinically curated real-world cohorts, pathology confirmation, and multidisciplinary adjudication for validation. Phase 1 (6 months) will prove feasibility and operational compatibility (governance, data workflows, and grounded outputs) using representative rare cancer cases and evidence sources. Phase 2 (2.5 years) will deliver full disease modules for prioritized selected rare

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	cancer indications, validate clinical utility (tumor board concordance, trial yield, and safety via abstention), and produce deployable modules and pharma-ready dossiers that support commercialization and external partnerships.
Key Words	1. Generative AI, LLM, Retrieval-augmented generation (RAG) 2. Rare cancers 3. Precision oncology decision support 4. Drug repurposing and rational combinations 5. Clinical trial matching and evidence grading

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	1) Develop and maintain disease-focused RAG libraries and an evidence graph for selected rare cancers under-represented in typical trials, with provenance and evidence tagging. 2) Build a constrained Q&A pipeline that generates patient-specific investigational therapy/combination options and trial matches with mandatory citations, evidence grading, uncertainty scoring, and abstention. 3) Create product-grade report templates and a transparent prioritization score that ranks options by mechanistic fit, evidence, safety/PK, feasibility, and biomarker suitability. 4) Clinically validate utility using institutional real-world cohorts and multidisciplinary adjudication, and deliver deployable rare caner modules plus cohort/target dossiers for downstream partnerships.			
R&D Target Specifications	• Grounding and safety: ≥95% citation coverage for all recommendation statements; ≤2% rate of unsupported claims on audited samples; abstention invoked when evidence is below predefined thresholds. • Retrieval quality: clinician-rated relevance ≥80% on a curated evaluation set; improved precision/coverage after feedback iterations. • Clinical utility (primary): tumor board appropriateness/concordance for top-3 options ≥80% (shadow-mode adjudication). • Trial matching (primary/secondary): ≥30% increase in high-quality trial matches identified vs standard manual workflow and ≥50% reduction in time-to-match in pilot workflow studies (benchmark time 1 week). • Deliverables: validated rare disease-specific modules integrated into the industry platform with versioning and performance dashboards.			
R&D Plan	Phase 1 (Months 0–6; Budget $76,000): Establish governance (DUA/IRB, security, operating cadence), ingest seed corpora spanning rare cancers (literature and trial records), implement a minimum viable RAG prototype with provenance and evidence tagging, generate 10–20 pilot shadow-mode reports, and conduct MTB-style review to refine retrieval, evidence rules, and abstention. Deliver a Go/No-Go package and lock Phase 2 scope and validation plan. Phase 2 (Months 7–36; Budget $1.02M): Expand RAG and the evidence graph to prioritized selected rare cancer coverage; implement the full Q&A pipeline with safety gates and an evaluation harness; integrate trial matching and prioritization scoring; generate disease playbooks plus cross-cutting personalization modules (epigenetic, DDR/replication stress, immune phenotype, pathway-state, angiogenic/stromal, cell-cycle, resistance patterns). Conduct shadow deployment and retrospective validation using institutional cohorts with multidisciplinary adjudication (primary endpoints: MTB appropriateness/concordance and trial-match yield). Lock v2 modules with dashboards, produce publications, and generate cohort/target dossiers and an external pilot plan.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Provide pathology-confirmed case identification and clinically curated real-world datasets across prioritized rare cancers, including molecular profiles, treatments, and outcomes; define clinical report templates and evidence grading requirements; lead multidisciplinary adjudication (tumor board review) to score appropriateness, safety, and usability; perform validation analyses with biostatistical oversight; ensure compliance with governance/IRB/DUA and contribute to dissemination (manuscripts, abstracts) and pilot workflow design.

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Expected R&R of Korean SME Partner	Provide and operate the AI platform core, including disease-focused RAG infrastructure, orchestration of the Q&A pipeline, prioritization and scoring logic, and product-grade report generation for patient-level outputs and cohort/target summaries. Implement essential safety and governance mechanisms such as citation requirements, evidence consistency checks, and abstention rules, along with basic MLOps capabilities for versioning and monitoring. Support integration of clinical trial matching and shadow-mode deployment, and package validated disease modules and summary dossiers to facilitate external pilot studies and downstream industry collaboration.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological outcomes: (i) validated rare disease modules within a grounded RAG + LLM pipeline, (ii) transparent evidence grading and abstention controls suitable for clinical decision support, (iii) automated trial matching and structured report outputs, and (iv) performance dashboards for monitoring grounding, safety, and utility. Clinical impact: improved access to rational investigational options and trials for rare cancer patients, enhanced tumor board efficiency, and more consistent evidence synthesis across rare subtypes. Commercialization and economic impact: a defensible, validated rare-cancer product module to enable external pilots and partnerships; pharma-ready cohort/target dossiers supporting basket trial concepts, label expansion strategies, and real-world evidence collaborations. Public support is justified because the project translates AI into measurable clinical utility in underserved rare cancers while strengthening an SME's globally scalable platform.

6. References

(1) Lewis P, Perez E, Piktus A, et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS. 2020.

(2) Cerami E, Gao J, Dogrusoz U, et al. The cBio Cancer Genomics Portal: An Open Platform for Exploring Multidimensional Cancer Genomics Data. Cancer Discovery. 2012.